

## Project Design Phase Solution Architecture

|               |                                                                                 |
|---------------|---------------------------------------------------------------------------------|
| Date          | 14 feb 2026                                                                     |
| Team ID       | LTVIP2026TMIDS35912                                                             |
| Project Name  | Deep Learning Fundus Image Analysis for Early Detection of Diabetic Retinopathy |
| Maximum Marks | 2 Marks                                                                         |

### Solution Architecture

This solution architecture describes the design of a deep learning-powered web application for the early detection of Diabetic Retinopathy using retinal (fundus) images. The system allows patients or healthcare professionals to upload retinal images through a secure and user-friendly web interface, enabling automated screening for diabetic eye disease.

The frontend interface communicates with a Django-based backend, which handles user authentication, image upload, and request processing. The backend integrates a trained Convolutional Neural Network (CNN) model built using PyTorch to analyze retinal images and classify the presence or absence of Diabetic Retinopathy.

The uploaded images undergo preprocessing before being passed to the deep learning model to ensure consistent and accurate predictions. The prediction results are generated in real time and displayed to the user, supporting early medical decision-making and timely consultation with eye specialists.

In addition to prediction functionality, the system includes secure user registration, login, and profile management, ensuring data privacy and personalized access. All trained model weights and preprocessing components are efficiently stored and loaded on demand, enabling scalability and smooth deployment of the application.

### Solution Architecture Diagram

Figure 1: Architecture and data flow of the *Deep Learning Fundus Image Analysis for Early Detection of Diabetic Retinopathy* application.



